Matrix Inverse

Definition. Let A be an $n \times n$ (square) matrix. We say that A is ***invertible*** if there is an $n \times n$ matrix B such that

$$AB = I_n \quad \text{and} \quad BA = I_n.$$

In this case, the matrix B is called the ***inverse*** of A, and we write $B = A^{-1}$.

Facts about invertible matrices. *Let A and B be invertible $n \times n$ matrices.*

1. *A^{-1} is invertible, and its inverse is $(A^{-1})^{-1} = A$.*
2. *AB is invertible, and its inverse is $(AB)^{-1} = B^{-1}A^{-1}$ (note the order).*

$(AB)^T = B^T A^T$

Definition. The ***determinant*** of a 2×2 matrix is the number

$$\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

Proposition. *Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.*

1. *If $\det(A) \neq 0$, then A is invertible, and*

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

2. *If $\det(A) = 0$, then A is not invertible.*

For general nxn matrix A
to find A^{-1} :

$[A \mid I_n] \xrightarrow{\text{Row Reduction}} [I_n \mid A^{-1}]$

Proposition.

1. *A transformation $T : \mathbf{R}^n \to \mathbf{R}^n$ is invertible if and only if it is both one-to-one and onto.*
2. *If T is already known to be invertible, then $U : \mathbf{R}^n \to \mathbf{R}^n$ is the inverse of T provided that either $T \circ U = \mathrm{Id}_{\mathbf{R}^n}$ or $U \circ T = \mathrm{Id}_{\mathbf{R}^n}$: it is only necessary to verify one.*

The Invertible Matrix Theorem

Invertible Matrix Theorem. *Let A be an $n \times n$ matrix, and let $T : \mathbf{R}^n \to \mathbf{R}^n$ be the matrix transformation $T(x) = Ax$. The following statements are equivalent:*

1. *A is invertible.*
2. *A has n pivots.*
3. *$\mathrm{Nul}(A) = \{0\}$.*
4. *The columns of A are linearly independent.*
5. *The columns of A span $\mathbf{R}^n$.*
6. *$Ax = b$ has a unique solution for each b in $\mathbf{R}^n$.*
7. *T is invertible.*
8. *T is one-to-one.*
9. *T is onto.*

Determinants: Definition

⚠ **Definition.** The ***determinant*** is a function

$$\det\colon \{\text{square matrices}\} \longrightarrow \mathbf{R}$$

satisfying the following properties:

1. Doing a row replacement on A does not change $\det(A)$.
2. Scaling a row of A by a scalar c multiplies the determinant by c.
3. Swapping two rows of a matrix multiplies the determinant by -1.
4. The determinant of the identity matrix I_n is equal to 1.

Definition. The ***determinant*** of a 2×2 matrix is the number

$$\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

Proposition. *Let A be an $n \times n$ matrix.*

1. *If A has a zero row or column, then $\det(A) = 0$.*
2. *If A is upper-triangular or lower-triangular, then $\det(A)$ is the product of its diagonal entries.*

upper-triangular

$$\begin{pmatrix} \star & \star & \star & \star \\ 0 & \star & \star & \star \\ 0 & 0 & \star & \star \\ 0 & 0 & 0 & \star \end{pmatrix}$$

lower-triangular

$$\begin{pmatrix} \star & 0 & 0 & 0 \\ \star & \star & 0 & 0 \\ \star & \star & \star & 0 \\ \star & \star & \star & \star \end{pmatrix}$$

diagonal

$$\begin{pmatrix} \star & 0 & 0 & 0 \\ 0 & \star & 0 & 0 \\ 0 & 0 & \star & 0 \\ 0 & 0 & 0 & \star \end{pmatrix}$$

Definition. The ***transpose*** of an $m \times n$ matrix A is the $n \times m$ matrix A^T whose rows are the columns of A. In other words, the ij entry of A^T is a_{ji}.

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{pmatrix} \longrightarrow A^T = \begin{pmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \\ a_{14} & a_{24} & a_{34} \end{pmatrix}$$

flip

Fact. *Let A be an $m \times n$ matrix, and let B be an $n \times p$ matrix. Then*

$$(AB)^T = B^T A^T.$$

properties of determinants

Invertibility Property. *A square matrix is invertible if and only if $\det(A) \neq 0$.*

Multiplicativity Property. *If A and B are $n \times n$ matrices, then*

$$\det(AB) = \det(A)\det(B).$$

Corollary. *If A is a square matrix, then*

$$\det(A^n) = \det(A)^n$$

for all $n \geq 1$. If A is invertible, then the equation holds for all $n \leq 0$ as well; in particular,

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

Transpose Property. *For any square matrix A, we have*

$$\det(A) = \det(A^T).$$

Cofactors Expansions

Definition. Let A be an $n \times n$ matrix.

1. The (i,j) ***minor,*** denoted A_{ij}, is the $(n-1)\times(n-1)$ matrix obtained from A by deleting the ith row and the jth column.
2. The (i,j) ***cofactor*** C_{ij} is defined in terms of the minor by

$$C_{ij} = (-1)^{i+j}\det(A_{ij}).$$

Theorem (Cofactor expansion). *Let A be an $n \times n$ matrix with entries a_{ij}.*

1. *For any $i = 1, 2, \ldots, n$, we have*

$$\det(A) = \sum_{j=1}^{n} a_{ij}C_{ij} = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}.$$

This is called ***cofactor expansion along the ith row.***

2. *For any $j = 1, 2, \ldots, n$, we have*

$$\det(A) = \sum_{i=1}^{n} a_{ij}C_{ij} = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}.$$

This is called ***cofactor expansion along the jth column.***

Example:

The determinant of a 3×3 matrix. We can also use cofactor expansions to find a formula for the determinant of a 3×3 matrix. Let is compute the determinant of

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

by expanding along the first row. The minors and cofactors are:

$$A_{11} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = \begin{pmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{pmatrix} \qquad C_{11} = +\det\begin{pmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{pmatrix}$$

$$A_{12} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = \begin{pmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{pmatrix} \qquad C_{12} = -\det\begin{pmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{pmatrix}$$

$$A_{13} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = \begin{pmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{pmatrix} \qquad C_{13} = +\det\begin{pmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{pmatrix}$$

The determinant is:

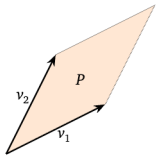
$$\begin{aligned} \det(A) &= a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} \\ &= a_{11}\det\begin{pmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{pmatrix} - a_{12}\det\begin{pmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{pmatrix} + a_{13}\det\begin{pmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{pmatrix} \\ &= a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}) \\ &= a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33}. \end{aligned}$$

Determinants and Volumes

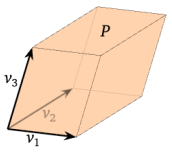
Definition. The ***paralellepiped*** determined by n vectors $v_1, v_2, \ldots, v_n$ in $\mathbf{R}^n$ is the subset

$$P = \{a_1x_1 + a_2x_2 + \cdots + a_nx_n \mid 0 \le a_1, a_2, \ldots, a_n \le 1\}.$$

Example (Parallelograms). When $n = 2$, a paralellepiped is just a paralellogram in $\mathbf{R}^2$. Note that the edges come in parallel pairs.



Example. When $n = 3$, a parallelepiped is a kind of a skewed cube. Note that the faces come in parallel pairs.



Theorem (Determinants and volumes). *Let $v_1, v_2, \ldots, v_n$ be vectors in $\mathbf{R}^n$, let P be the parallelepiped determined by these vectors, and let A be the matrix with rows $v_1, v_2, \ldots, v_n$. Then the absolute value of the determinant of A is the volume of P:*

$$|\det(A)| = \mathrm{vol}(P).$$